LaBRADOR

October 29, 2026

Blockstream Research

research.blockstream.com

github.com/BlockstreamResearch/zk-book



Plan

1 Dot Product Constraint Systems

2 The Main Protocol

3 Recursion

4 Parameters and Security

5 Reduction From R1CS

What will we learn today?

- ALS20/LNP22 proofs grow **linearly** with the witness (10–20 KB for small statements). For circuits with a million gates, this is no better than sending the witness.
- **LaBRADOR** (*Lattice-Based Recursively Amortized Demonstration Of R1CS*): an R1CS with 2^{20} constraints in about 58 KB, based only on Module-SIS.
- Two ideas:
  - **amortization**: fold r secret vectors into one random linear combination;
  - **recursion** (as in Bulletproofs): the last message is a witness for a statement of the same form.

Key idea

Whenever the last message of a proof is a solution to a system of the same type as the original one, the prover can recursively prove knowledge of it instead of sending it.

Before We Start

LaBRADOR is not zero-knowledge

This makes it noticeably simpler: no masks $\mathbf{y}$ and no rejection sampling.

Notation

- $\mathcal{R}_q = \mathbb{Z}_q[T]/(T^d + 1)$, where $T^d + 1$ splits into two irreducible factors modulo q (as in LNP22).
- $f^* := f(T^{-1})$ is the conjugate, $\mathrm{ct}(f)$ the constant coefficient.
- $\|\mathbf{f}\|_2$ is the ℓ^2 norm of the coefficient vector of $\mathbf{f} \in \mathcal{R}_q^n$; $\vec{\mathbf{f}} \in \mathbb{Z}_q^{nd}$ is the coefficient vector itself.

Plan: define dot product constraint systems → build the main protocol (Attempts #1–#4) → recursion → parameters → reduction from R1CS.

Dot Product Constraint Systems

Dot Product Functions

Requirements for the relation:

1. expressive enough to encode R1CS;
2. the verifier's final checks form a statement of the **same type** (recursion).

For $\mathbf{a}, \mathbf{b} \in \mathcal{R}_q^n$, $\langle \mathbf{a}, \mathbf{b} \rangle := \sum_{i \in [n]} a_i b_i \in \mathcal{R}_q$. A **dot product function** of $\mathbf{s}_1, \ldots, \mathbf{s}_r \in \mathcal{R}_q^n$ is

$$f(\mathbf{s}_1, \ldots, \mathbf{s}_r) = \sum_{i,j \in [r]} a_{ij} \langle \mathbf{s}_i, \mathbf{s}_j \rangle + \sum_{i \in [r]} \langle \boldsymbol{\varphi}_i, \mathbf{s}_i \rangle + a_0,$$

where $a_{ij}, a_0 \in \mathcal{R}_q$ and $\boldsymbol{\varphi}_i \in \mathcal{R}_q^n$ are public; w.l.o.g. $\mathbf{A} := (a_{ij})_{i,j}$ is symmetric.

Remark

A less general version of the quadratic forms $\mathbf{s}^\top \mathbf{Q} \mathbf{s} + \mathbf{p}^\top \mathbf{s} + p_0$ from LNP22. We denote this class of functions by $\mathcal{Q}$.

The Principal Relation

Definition (Principal Relation)

Fix the *rank* $n \geq 1$, the *multiplicity* $r \geq 1$, and a norm bound $\beta > 0$. The **principal relation** of LaBRADOR is

$$\mathcal{R}_{\mathbf{dot}} := \left\{ \begin{array}{l} \mathbb{x} = (\mathcal{F}, \mathcal{F}_{\mathsf{ct}}, \beta) \\ \mathbb{w} = (\mathbf{s}_1, \ldots, \mathbf{s}_r) \end{array} \middle| \begin{array}{l} f(\mathbf{s}_1, \ldots, \mathbf{s}_r) = 0 \text{ for all } f \in \mathcal{F}, \\ \mathsf{ct}(g(\mathsf{s}_1, \ldots, \mathsf{s}_r)) = 0 \text{ for all } g \in \mathcal{F}_{\mathsf{ct}}, \\ \sum_{i \in [r]} \|\mathbf{s}_i\|_2^2 \leq \beta^2 \end{array} \right\},$$

where $\mathcal{F}, \mathcal{F}_{\mathsf{ct}} \subseteq Q$ and $\mathbf{s}_1, \ldots, \mathbf{s}_r \in \mathcal{R}_q^n$.

- $\mathcal{F}$: constraints that vanish as **whole polynomials**.
- $\mathcal{F}_{\mathsf{ct}}$: only the **constant coefficient** must vanish; by the constant coefficient lemma, these are linear and quadratic relations over $\mathbb{Z}_q$ among the coefficients.

Examples

Example

Statements on a single vector $\mathbf{s} \in \mathcal{R}_q^n$ $(r = 1)$:

1. *Opening of an Ajtai commitment.* $\mathbf{t} = \mathbf{A}\mathbf{s}$ for $\mathbf{A} \in \mathcal{R}_q^{\kappa \times n}$ with rows $\mathbf{a}_1^\top, \ldots, \mathbf{a}_\kappa^\top$ is the collection of κ linear functions $f_k(\mathbf{s}) = \langle \mathbf{a}_k, \mathbf{s} \rangle - t_k$ in $\mathcal{F}$.

2. *A single coefficient.* The j-th coefficient of the i-th entry of $\mathbf{s}$ equals $\mathrm{ct}(\langle (T^j \mathbf{e}_i)^*, \mathbf{s} \rangle)$. "This coefficient equals 5" is $g(\mathbf{s}) = \langle (T^j \mathbf{e}_i)^*, \mathbf{s} \rangle - 5$ in $\mathcal{F}_{\mathsf{ct}}$.

3. *Norm.* $\|\mathbf{s}\|_2^2 = \mathrm{ct}(\langle \mathbf{s}^*, \mathbf{s} \rangle)$ is not a dot product function of $\mathbf{s}$ alone (the conjugate appears). Fix: add $\mathbf{s}^*$ to the witness (later). The bound β is already part of the relation.

Question #1

Write "I know $(\mathbf{s}, \mathbf{e})$ with $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$ and $\|\mathbf{s}\|_2^2 + \|\mathbf{e}\|_2^2 \leq \beta^2$" (a Module-LWE key) as an instance of $\mathcal{R}_{\mathsf{dot}}$ with $r = 1$.

The Main Protocol

Attempt #1: Amortized Opening

The trivial protocol sends $\mathbf{s}_1, \ldots, \mathbf{s}_r$: rn ring elements. Commit $\mathbf{t}_i = \mathbf{A}\mathbf{s}_i$ for $\mathbf{A} \in \mathcal{R}_q^{\kappa \times n}$. No zero-knowledge $\Rightarrow$ drop the mask; one response for all commitments:

$$\mathbf{z} := c_1\mathbf{s}_1 + \cdots + c_r\mathbf{s}_r.$$

$\mathcal{P}(\mathbb{x} = (\mathbf{t}_i)_{i\in[r]}, \mathbb{w} = (\mathbf{s}_i)_{i\in[r]})$:

1: **receive** $c_1, \ldots, c_r$;
2: *// amortized opening*
send $\mathbf{z} \leftarrow \sum_{i\in[r]} c_i\mathbf{s}_i$.

$\mathcal{V}(\mathbb{x} = (\mathbf{t}_i)_{i\in[r]}) \to \{0, 1\}$:

1: **send** $c_1, \ldots, c_r \leftarrow\$ C$;
2: **receive** $\mathbf{z}$;
3: **assert** $\|\mathbf{z}\|_2 \le \gamma$;
4: **return** $[\![\mathbf{A}\mathbf{z} = \sum_{i\in[r]} c_i\mathbf{t}_i]\!]$.

- Communication: $r\kappa + n$ instead of rn ring elements (κ is a few dozen).
- Price: coefficients of $\mathbf{z}$ grow by $\approx \sqrt{r\tau}$, where $\tau := \max_{c\in C} \|c\|_2^2$.

The Challenge Set

The extractor gets a *product* of an extracted vector with a challenge difference; bounding it needs the **operator norm**

$$\|c\|_{\mathsf{op}} := \sup_{f \in \mathcal{R} \setminus \{0\}} \frac{\|cf\|_2}{\|f\|_2}.$$

LaBRADOR requires: $c - c'$ invertible in $\mathcal{R}_q$ for distinct challenges, $\|c\|_2^2 \leq \tau$, and $\|c\|_{\mathsf{op}} \leq \theta$.

With $N_i(c) := \#\{j : |c_j| = i\}$ and $k_0 + k_1 + k_2 = d$:

$$C := \{c \in \mathcal{R} : N_0(c) = k_0,\ N_1(c) = k_1,\ N_2(c) = k_2,\ \|c\|_{\mathsf{op}} \leq \theta\},$$

with $d = 64$, $(k_0, k_1, k_2) = (23, 31, 10)$, $\theta = 15$ (≈ 6 candidates per challenge).

Question #2

Express $\tau = \|c\|_2^2$ (for $c \in C$) and $\#C$ in terms of d, k_0, k_1, k_2.

Soundness and Weak Openings

As with the Lyubashevsky protocol, the extractor does not obtain the $\mathbf{s}_i$ themselves.

Definition (Weak opening)

Vectors $\mathbf{s}_i^*$ with $\mathbf{A}\mathbf{s}_i^* = \mathbf{t}_i$, together with a challenge difference $\bar{c} = c - c'$ such that $\bar{c}\mathbf{s}_i^*$ is short.

- The commitment stays binding for weak openings under Module-SIS, with a norm bound larger by a factor proportional to θ.
- Two weak openings $(\mathbf{s}^*, \bar{c})$, $(\mathbf{s}'^*, \bar{c}')$ of the same commitment: $\bar{c}'(\bar{c}\mathbf{s}^*) - \bar{c}(\bar{c}'\mathbf{s}'^*)$ lies in the kernel of $\mathbf{A}$, and its norm is bounded using the operator norms.

That is why the operator norm of the challenges enters the parameters.

Attempt #2: Garbage Polynomials

Goal: show $F(\mathbf{s}_1, \ldots, \mathbf{s}_r) = 0$ for a single dot product function F. As in ALS20, evaluate on $\mathbf{z} = \sum_i c_i \mathbf{s}_i$:

$$\langle \mathbf{z}, \mathbf{z} \rangle = \sum_{i,j \in [r]} g_{ij}\, c_i c_j, \qquad \sum_{i \in [r]} \langle \boldsymbol{\varphi}_i, \mathbf{z} \rangle c_i = \sum_{i,j \in [r]} h_{ij}\, c_i c_j,$$

where the **garbage polynomials** are

$$g_{ij} := \langle \mathbf{s}_i, \mathbf{s}_j \rangle, \qquad h_{ij} := \frac{1}{2}(\langle \boldsymbol{\varphi}_i, \mathbf{s}_j \rangle + \langle \boldsymbol{\varphi}_j, \mathbf{s}_i \rangle).$$

Both matrices are symmetric and **do not depend on the challenges**.

Moreover, the constraint is a linear function of the garbage:

$$F(\mathbf{s}_1, \ldots, \mathbf{s}_r) = \sum_{i,j \in [r]} a_{ij} g_{ij} + \sum_{i \in [r]} h_{ii} + a_0.$$

So: send the garbage together with $\mathbf{z}$, and check the three equations.

Attempt #2: the Protocol

$\mathcal{P}(\mathbb{x}, \mathbb{w} = (\mathbf{s}_i)_{i\in[r]})$:

1: $g_{ij} \leftarrow \langle \mathbf{s}_i, \mathbf{s}_j \rangle$ for $i \le j$;
2: $h_{ij} \leftarrow \frac{1}{2}(\langle \boldsymbol{\varphi}_i, \mathbf{s}_j \rangle + \langle \boldsymbol{\varphi}_j, \mathbf{s}_i \rangle)$ for $i \le j$;
3: **send** $(g_{ij}, h_{ij})_{i\le j}$;
4: **receive** $c_1, \ldots, c_r$;
5: **send** $\mathbf{z} \leftarrow \sum_{i\in[r]} c_i \mathbf{s}_i$.

$\mathcal{V}(\mathbb{x} = ((\mathbf{t}_i)_{i\in[r]}, F)) \to \{0, 1\}$:

1: **receive** $(g_{ij}, h_{ij})_{i\le j}$;
2: **send** $c_1, \ldots, c_r \leftarrow\$ C$;
3: **receive** $\mathbf{z}$;
4: **assert** $\|\mathbf{z}\|_2 \le \gamma$;
5: **assert** $\mathbf{A}\mathbf{z} = \sum_i c_i \mathbf{t}_i$;
6: **assert** $\langle \mathbf{z}, \mathbf{z} \rangle = \sum_{i,j} g_{ij} c_i c_j$;
7: **assert** $\sum_i \langle \boldsymbol{\varphi}_i, \mathbf{z} \rangle c_i = \sum_{i,j} h_{ij} c_i c_j$;
8: **return** $[\![\sum_{i,j} a_{ij} g_{ij} + \sum_i h_{ii} + a_0 = 0]\!]$.

Why sound? The garbage is sent *before* the challenges: a lie makes the two sides of a check different quadratic polynomials in $c_1, \ldots, c_r$, agreeing on random challenges with negligible probability (Schwartz–Zippel over $\mathcal{R}_q$).

Price: $r^2 + r$ garbage polynomials, quadratic in r (we return to this later).

Attempt #3: Aggregating Constraints

Aggregating $\mathcal{F}$. For $f_1, \ldots, f_K \in \mathcal{F}$, the verifier sends $\alpha_1, \ldots, \alpha_K \leftarrow\$ \mathcal{R}_q$, and the prover proves that $F := \sum_{k\in[K]} \alpha_k f_k$ vanishes (again a dot product function).

Aggregating $\mathcal{F}_{\mathsf{ct}}$. Coefficients must come from $\mathbb{Z}_q$ (multiplying by a general element of $\mathcal{R}_q$ may break $\mathsf{ct} = 0$). One combination has error $1/q$, so take $K_{\mathsf{ct}} := \lceil \lambda / \log q \rceil$ combinations with $\boldsymbol{\psi}_1, \ldots, \boldsymbol{\psi}_{K_{\mathsf{ct}}} \leftarrow\$ \mathbb{Z}_q^L$:

$$G_k := \sum_{\ell\in[L]} \psi_{k,\ell} g_\ell, \quad k \in [K_{\mathsf{ct}}].$$

For $q \approx 2^{32}$, only $K_{\mathsf{ct}} = 4$.

Attempt #3: Constant Coefficients without Masking

No masking needed

Write $G_k = Q_k + b_k$. LNP22 had to mask the non-constant coefficients; LaBRADOR sends $v_k := Q_k(\mathbf{s}_1, \ldots, \mathbf{s}_r)$ in the clear. The verifier checks $\mathrm{ct}(v_k) = -\mathrm{ct}(b_k)$, and

$$Q_k(\mathbf{s}_1, \ldots, \mathbf{s}_r) - v_k = 0$$

is an *ordinary* constraint added to $\mathcal{F}$ before aggregating it.

Result

The whole statement collapses into a single function F, proven with Attempt #2 at the cost of only K_{ct} additional polynomials v_k.

Attempt #4: Johnson–Lindenstrauss Projection

$\|\mathbf{z}\|_2 \le \gamma$ only yields weak openings (norm may exceed β a lot); the R1CS reduction needs a small slack.

With $\mathcal{B}$ on $\{0, \pm 1\}$, $\Pr[0] = 1/2$, $\Pr[\pm 1] = 1/4$: the verifier sends $\mathbf{\Pi}_1, \ldots, \mathbf{\Pi}_r \leftarrow\$ \mathcal{B}^{256 \times nd}$, the prover sends

$$\mathbf{p} := \sum_{i \in [r]} \mathbf{\Pi}_i \vec{\mathbf{s}}_i \in \mathbb{Z}_q^{256},$$

and the verifier checks that $\|\mathbf{p}\|_2 \le \sqrt{128}\beta$.

Lemma

Let $b \le q/125$ and let $\mathbf{w} \in \mathbb{Z}_q^m$ be any vector with $\|\mathbf{w}\|_2 \ge b$. Then,

$$\Pr_{\mathbf{\Pi} \leftarrow\$ \mathcal{B}^{256 \times m}} \left[\|\mathbf{\Pi}\mathbf{w} \bmod q\|_2 < \sqrt{30}\, b \right] \lesssim 2^{-128}.$$

A projection of a long vector is long, even after the reduction modulo q.

Attempt #4: Analysis

- Honest case: $\mathbb{E}[\|\mathbf{\Pi w}\|_2^2] = 128\|\mathbf{w}\|_2^2$.
- For $\mathbf{w} = (\vec{\mathbf{s}}_1, \ldots, \vec{\mathbf{s}}_r)$: if $\|\mathbf{p}\|_2 \leq \sqrt{128}\beta$, then $\sum_i \|\mathbf{s}_i\|_2^2 \leq \frac{128}{30}\beta^2$ except w.p. 2^{-128}: slack $\sqrt{128/30} \approx 2.07$.
- The honest $\mathbf{p}$ passes only w.p. $\approx 1/2$: the prover asks for new matrices (affects only the running time).

Correctness of $\mathbf{p}$ for free. Let $\boldsymbol{\pi}_{i,j} \in \mathcal{R}_q^n$ have the j-th row of $\mathbf{\Pi}_i$ as coefficient vector. Then

$$\mathbf{ct}\Big(\sum_{i\in[r]} \langle \boldsymbol{\pi}_{i,j}^*, \mathbf{s}_i \rangle - p_j\Big) = 0$$

is a constant-coefficient constraint: add these 256 functions to $\mathcal{F}_{\mathsf{ct}}$.

Question #3

Let $\mathbf{w} \in \mathbb{Z}^m$ and $\boldsymbol{\pi} \leftarrow\!\$\ \mathcal{B}^m$. Show that $\mathbb{E}[\langle \boldsymbol{\pi}, \mathbf{w} \rangle^2] = \frac{1}{2}\|\mathbf{w}\|_2^2$, and deduce the expectation above.

The Base Protocol: Message Flow

Prover $\mathcal{P}$		Verifier $\mathcal{V}$
commit: $\mathbf{t}_i \leftarrow \mathbf{A}\mathbf{s}_i$, $i \in [r]$	$\xrightarrow{(\mathbf{t}_i)_i}$	
	$\xleftarrow{(\mathbf{\Pi}_i)_i}$	$\mathbf{\Pi}_i \leftarrow\$ \mathcal{B}^{256\times nd}$
	$\xrightarrow{\mathbf{p}}$	**assert** $\|\mathbf{p}\|_2 \leq \sqrt{128}\beta$
project: $\mathbf{p} \leftarrow \sum_i \mathbf{\Pi}_i \vec{\mathbf{s}}_i$		
	$\xleftarrow{(\boldsymbol{\psi}_k)_k}$	$\boldsymbol{\psi}_k \leftarrow\$ \mathbb{Z}_q^{L+256}$
	$\xrightarrow{(v_k)_k}$	**assert** $\mathrm{ct}(v_k) = -\mathrm{ct}(b_k)$
aggregate: $v_k \leftarrow Q_k(\mathbf{s}_1, \ldots, \mathbf{s}_r)$		
	$\xleftarrow{\boldsymbol{\alpha}}$	$\boldsymbol{\alpha} \leftarrow\$ \mathcal{R}_q^{K+K_{\mathrm{ct}}}$
	$\xrightarrow{(g_{ij},h_{ij})_{i\leq j}}$	
aggregate into F; garbage g_{ij}, h_{ij}		
	$\xleftarrow{c_1,\ldots,c_r}$	$c_i \leftarrow\$ C$
	$\xrightarrow{\mathbf{z}}$	**assert** the checks of Attempt #2
amortize: $\mathbf{z} \leftarrow \sum_i c_i \mathbf{s}_i$		

Order matters: the projection is aggregated, so it comes before $\boldsymbol{\psi}_k$; h_{ij} depend on the aggregated $\boldsymbol{\varphi}_i$, so they come after $\boldsymbol{\alpha}$.

Recursion

The Base Protocol Is Not Succinct

Its proof contains $r\kappa$ commitments, $r^2 + r$ garbage polynomials, and $\mathbf{z}$ of rank n.

Observation

All checks on the last message are **dot product constraints**:

- $\mathbf{Az} = \sum_i c_i \mathbf{t}_i$ is linear in $\mathbf{z}$ and $\mathbf{t}_i$;
- $\langle \mathbf{z}, \mathbf{z} \rangle = \sum_{i,j} g_{ij} c_i c_j$ is quadratic in $\mathbf{z}$ and linear in g_{ij};
- the remaining two equations are linear in $\mathbf{z}$, g_{ij}, h_{ij}.

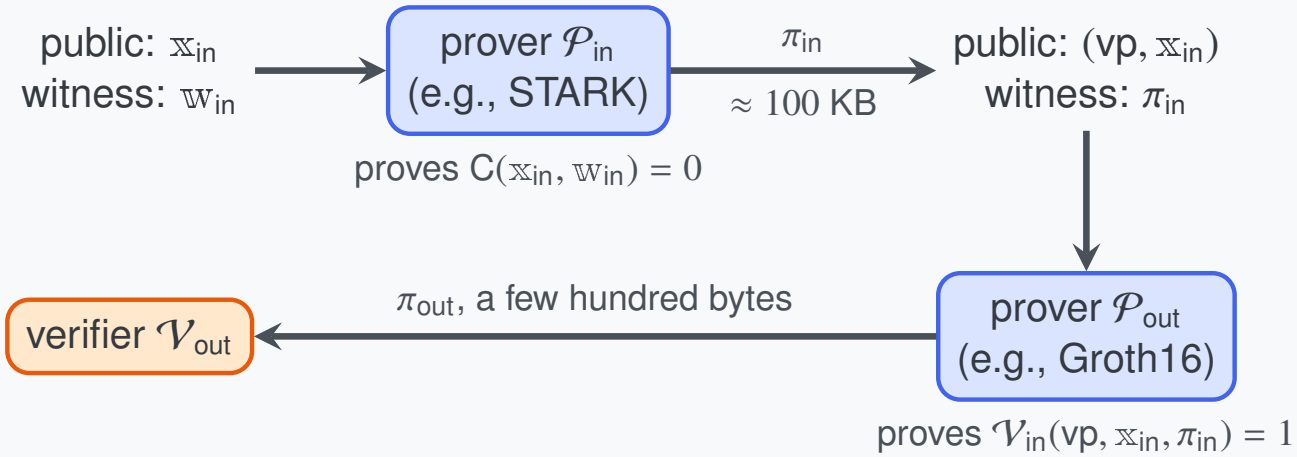
All coefficients (c_i, $\mathbf{A}$, a_{ij}, $\boldsymbol{\varphi}_i$, a_0) are known to the verifier.

The last message is a witness for a *new* instance of $\mathcal{R}_{\text{dot}}$: prove knowledge of it by running the protocol again!

Warm-up: Proof Compression

Fast prover with large proofs (STARK, ≈ 100 KB) + slow prover with tiny proofs (Groth16). Prove "I know an accepting inner proof":

$$\mathcal{R}_{\text{out}} := \{(\mathbb{x}_{\text{out}} := (\text{vp}, \mathbb{x}_{\text{in}}), \mathbb{w}_{\text{out}} := \pi_{\text{in}}) : \mathcal{V}_{\text{in}}(\text{vp}, \mathbb{x}_{\text{in}}, \pi_{\text{in}}) = 1\}.$$



The slow prover handles only the (small) verification circuit of $\mathcal{V}_{\text{in}}$. Used in practice to post STARK proofs on-chain.

Knowledge Soundness of the Composition

Proposition

Let κ_{in} and κ_{out} be the knowledge errors of inner and outer systems. The resultant proving system has a knowledge error of $\kappa_{\text{in}} + \kappa_{\text{out}}$.

Proof idea. Let $\mathcal{A}$ convince $\mathcal{V}_{\text{out}}$ with probability ϵ.

1: $\mathcal{E}_{\text{out}}$ obtains an accepting π_{in} w.p. $\geq \epsilon - \kappa_{\text{out}}$;
2: so $\mathcal{E}_{\text{out}}$ (run on $\mathcal{A}$) is a successful inner prover;
3: $\mathcal{E}_{\text{in}}$, run on $\mathcal{E}_{\text{out}}$, obtains $\mathbb{w}$ w.p. $\geq \epsilon - (\kappa_{\text{out}} + \kappa_{\text{in}})$. ■

Caveats

1. Extractors are *nested*: k levels may cost 2^k times longer, so only a few levels (at most logarithmic in λ).
2. With Fiat-Shamir inside, the circuit contains a concrete hash function: a heuristic assumption.

Reductions of Knowledge

An interactive protocol may not decide whether to accept, but *reduce* a claim about one relation to a claim about another. A relation $\mathcal{R}$ consists of triples $(\mathsf{pp}, \mathbb{x}, \mathbb{w})$.

Definition (Reduction of Knowledge)

A **reduction of knowledge** Π from $\mathcal{R}_1$ to $\mathcal{R}_2$ consists of

- $\mathsf{Setup}(1^\lambda) \to \mathsf{pp}$;
- $\langle \mathcal{P}(\mathsf{pp}, \mathbb{x}_1, \mathbb{w}_1), \mathcal{V}(\mathsf{pp}, \mathbb{x}_1) \rangle \to (\mathbb{x}_2, \mathbb{w}_2)$: the verifier outputs a new statement $\mathbb{x}_2$ (or $\perp$), the prover a new witness $\mathbb{w}_2$.

- **Completeness**: honest $(\mathsf{pp}, \mathbb{x}_1, \mathbb{w}_1) \in \mathcal{R}_1$ gives $(\mathsf{pp}, \mathbb{x}_2, \mathbb{w}_2) \in \mathcal{R}_2$.
- **Knowledge soundness** (error κ): if $\mathcal{P}^*$ ends in $\mathcal{R}_2$ w.p. $\epsilon > \kappa$, an expected poly-time extractor outputs $\mathbb{w}_1$ with $(\mathsf{pp}, \mathbb{x}_1, \mathbb{w}_1) \in \mathcal{R}_1$ w.p. $\geq \epsilon - \kappa$.
- **Public reducibility**: $\mathbb{x}_2 = \mathsf{f}(\mathsf{pp}, \mathbb{x}_1, \mathsf{tr})$ for a deterministic poly-time f.

Knowledge Soundness, Formally

There exists an expected polynomial-time $\mathcal{E}$ such that for every expected polynomial-time $(\mathcal{A}, \mathcal{P}^*)$, if

$$\Pr\left[(\mathsf{pp}, \mathbb{x}_2, \mathbb{w}_2) \in \mathcal{R}_2 \;\middle|\; \begin{array}{l} \mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda), \\ (\mathbb{x}_1, \mathsf{st}) \leftarrow \mathcal{A}(\mathsf{pp}), \\ (\mathbb{x}_2, \mathbb{w}_2) \leftarrow \langle \mathcal{P}^*(\mathsf{st}), \mathcal{V}(\mathsf{pp}, \mathbb{x}_1) \rangle \end{array}\right] = \epsilon > \kappa,$$

then $\mathcal{E}^{\mathcal{A}, \mathcal{P}^*}(\mathsf{pp}, \mathbb{x}_1)$ outputs $\mathbb{w}_1$ with $(\mathsf{pp}, \mathbb{x}_1, \mathbb{w}_1) \in \mathcal{R}_1$ with probability at least $\epsilon - \kappa$.

In words

Whoever can produce a witness for the new statement $\mathbb{x}_2$ must have known a witness for the original statement $\mathbb{x}_1$.

Examples and Composition

Example

1. An argument of knowledge for $\mathcal{R}$ = a reduction to $\mathcal{R}_\top := \{(\mathsf{pp}, \top, \bot)\}$ ($\top$ means "accept").
2. Proof compression: $\mathcal{R}_{\mathsf{in}} \to \mathcal{R}_{\mathsf{out}} \to \mathcal{R}_\top$.
3. Π_{send}: send $\mathbb{w}$, output $\top$ iff $(\mathsf{pp}, \mathbb{x}, \mathbb{w}) \in \mathcal{R}$. Knowledge error 0, not succinct.

Theorem (Composition)

Let Π be a reduction of knowledge from $\mathcal{R}$ to $\mathcal{R}'$ with knowledge error κ.

(i) **Sequential.** If Π' reduces $\mathcal{R}'$ to $\mathcal{R}''$ with error κ', then $\Pi' \circ \Pi$ reduces $\mathcal{R}$ to $\mathcal{R}''$ with error $\kappa + \kappa'$.

(ii) **Parallel.** If $\widetilde{\Pi}$ reduces $\widetilde{\mathcal{R}}$ to $\widetilde{\mathcal{R}}'$, then $\Pi \times \widetilde{\Pi}$ reduces $\mathcal{R} \times \widetilde{\mathcal{R}}$ to $\mathcal{R}' \times \widetilde{\mathcal{R}}'$.

Proof of *(i)*: exactly the nested extraction of the warm-up.

LaBRADOR as a Chain of Reductions

Relaxed relations

Lattice reductions are complete for one relation but knowledge sound only for a *relaxed* one: e.g., complete for norm $\leq \beta$, extraction only guarantees $\leq \sqrt{128/30}\beta$. LaBRADOR chooses the Module-SIS parameters of each iteration with the slack of the next ones in mind.

Each iteration is $\Pi_{\mathsf{main}} : \mathcal{R}_{\mathsf{dot}} \to \mathcal{R}'_{\mathsf{dot}}$ (new parameters), and $\Pi_{\mathsf{R1CS}} : \mathcal{R}_{\mathsf{R1CS}} \to \mathcal{R}_{\mathsf{dot}}$. The complete protocol is

$$\Pi_{\mathsf{LaBRADOR}} := \Pi_{\mathsf{send}} \circ \underbrace{\Pi_{\mathsf{main}} \circ \cdots \circ \Pi_{\mathsf{main}}}_{\ell \text{ times}} \circ \Pi_{\mathsf{R1CS}},$$

with knowledge error $\kappa_{\mathsf{R1CS}} + \ell \cdot \kappa_{\mathsf{main}}$.

- Same idea as folding schemes (Nova, LatticeFold).
- Both caveats disappear: $O(\log \log N)$ iterations, and Fiat-Shamir is applied once to the whole public-coin protocol.

Three Issues

To turn the base protocol into a reduction from $\mathcal{R}_{\text{dot}}$ to itself:

- *Issue #1:* the commitments $\mathbf{t}_i$ and the garbage are sent in the clear, and their size grows with r;
- *Issue #2:* the coefficients of $\mathbf{z}$ are larger by $\approx \sqrt{r\tau}$: they would blow up after a few iterations;
- *Issue #3:* the new witness is one long $\mathbf{z}$ and several short vectors, while the protocol prefers many vectors of the same rank.

Outer commitments (Issue #1)

Move commitments and garbage into the new witness: the verifier need not *see* them, they just must be fixed before the challenges. Send a single Ajtai commitment to all **inner commitments** $\mathbf{t}_i$: an **outer commitment**.

Outer Commitments

Ajtai needs short messages, but $\mathbf{t}_i$ has arbitrary coefficients. Decompose in a small base δ:

$$\mathbf{t}_i = \sum_{k \in [t]} \mathbf{t}_{i,k} \delta^k, \quad \|\mathbf{t}_{i,k}\|_\infty \leq \frac{1}{2}\delta, \quad t = \lceil \log q / \log \delta \rceil;$$

similarly h_{ij} in base δ and the (shorter) g_{ij} in base δ'. Collect the parts into $\mathbf{t}$, $\mathbf{g}$, $\mathbf{h}$.

The prover sends two outer commitments

$$\mathbf{u}' := \mathbf{B}\mathbf{t} + \mathbf{C}\mathbf{g} \in \mathcal{R}_q^{\kappa'}, \qquad \mathbf{u}'' := \mathbf{D}\mathbf{h} \in \mathcal{R}_q^{\kappa''}.$$

- Two of them, as they are sent at different times: $\mathbf{t}_i$ and g_{ij} depend only on the witness (first message); h_{ij} depend on $\boldsymbol{\varphi}_i$ (after aggregation).
- Both equations are linear: $\kappa' + \kappa''$ additional constraints of the next instance.

Decomposing and Reshaping

Issue #2. Decompose $\mathbf{z} = \mathbf{z}_{\mathsf{low}} + \delta'' \mathbf{z}_{\mathsf{high}}$ with $\|\mathbf{z}_{\mathsf{low}}\|_\infty \leq \frac{1}{2}\delta''$; the quadratic check becomes

$$\langle \mathbf{z}_{\mathsf{low}}, \mathbf{z}_{\mathsf{low}} \rangle + 2\delta'' \langle \mathbf{z}_{\mathsf{low}}, \mathbf{z}_{\mathsf{high}} \rangle + \delta''^2 \langle \mathbf{z}_{\mathsf{high}}, \mathbf{z}_{\mathsf{high}} \rangle = \sum_{i,j} g_{ij} c_i c_j,$$

still a dot product constraint. Choose $\delta \approx \delta' \approx \delta''$: all parts about equally wide.

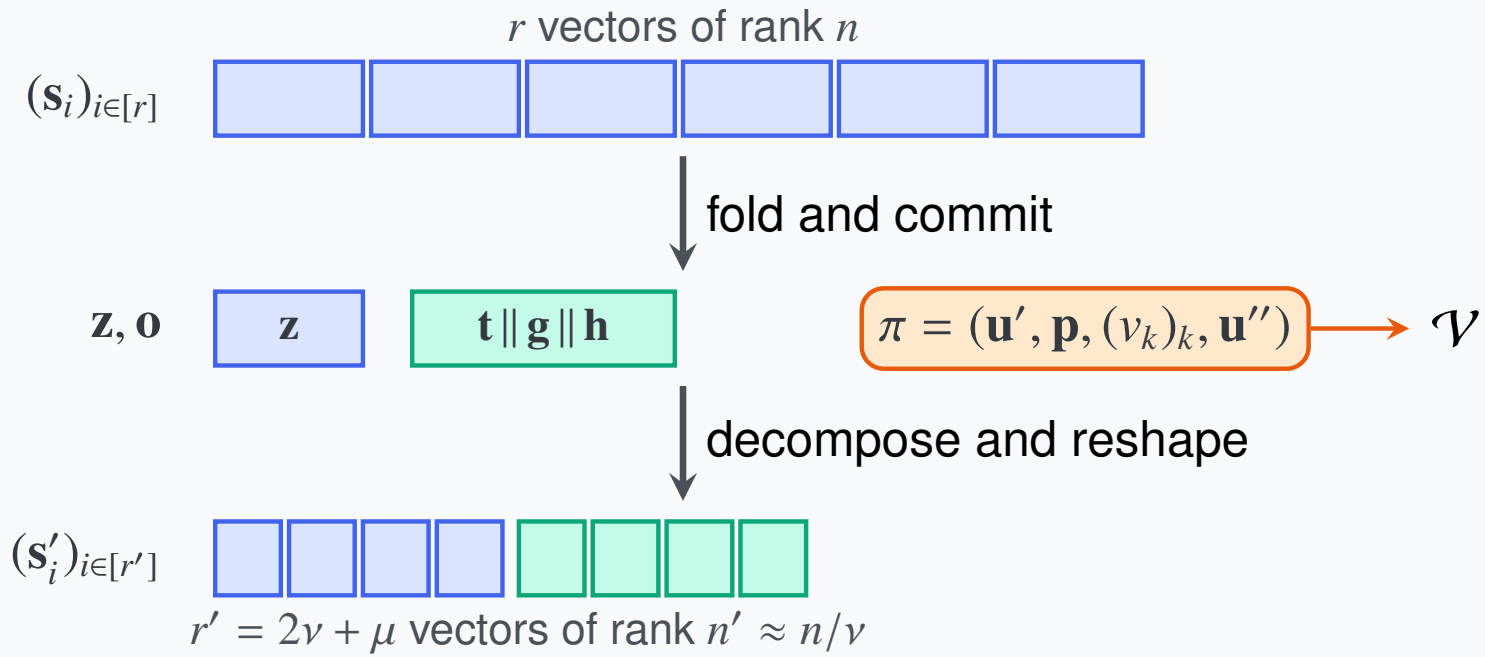
Issue #3. Let $\mathbf{o} := \mathbf{t} \,\|\, \mathbf{g} \,\|\, \mathbf{h} \in \mathcal{R}_q^m$. Split $\mathbf{z}_{\mathsf{low}}, \mathbf{z}_{\mathsf{high}}$ into ν pieces and $\mathbf{o}$ into μ pieces, pad to rank $n' := \max\{\lceil n/\nu \rceil, \lceil m/\mu \rceil\}$: a witness of multiplicity $r' = 2\nu + \mu$ ($n/\nu \approx m/\mu$).

The norm checks combine into one:

$$\sum_{i \in [r']} \|\mathbf{s}'_i\|_2^2 = \|\mathbf{z}_{\mathsf{low}}\|_2^2 + \|\mathbf{z}_{\mathsf{high}}\|_2^2 + \|\mathbf{o}\|_2^2 \leq \beta'^2.$$

New instance: $\kappa + \kappa' + \kappa'' + 3$ constraints in $\mathcal{F}$, empty $\mathcal{F}_{\mathsf{ct}}$, bound β'.

One Iteration of the Main Protocol



Instead of sending its last message, the prover outputs the new witness $(\mathbf{s}'_i)_{i\in[r']}$; the verifier outputs the new statement $(\mathcal{F}', \varnothing, \beta')$: $\mathbf{Az} = \sum_i c_i \mathbf{t}_i$, $\mathbf{u}' = \mathbf{Bt} + \mathbf{Cg}$, $\mathbf{u}'' = \mathbf{Dh}$, and the three garbage equations, rewritten in terms of the new witness.

How Fast Does the Witness Shrink?

Count ring elements, ignoring small constants:

- before: $N = rn$;
- after: $\approx 2n$ (for $\mathbf{z}_{\text{low}}$, $\mathbf{z}_{\text{high}}$) + $\approx r^2$ (for $\mathbf{o}$, dominated by garbage).

Minimizing $2N/r + r^2$ gives $r \approx N^{1/3}$ and a new witness of size $\approx N^{2/3}$.

$$N \to N^{2/3} \to N^{4/9} \to \ldots \quad \Rightarrow \quad O(\log \log N) \text{ iterations}$$

In practice, six or seven iterations; then the prover sends the remaining witness in the clear.

- Each iteration contributes only π: $\mathbf{u}', \mathbf{u}''$ ($\kappa' + \kappa''$ ring elements), $\mathbf{p}$ (256 small integers), K_{ct} polynomials v_k: a few kilobytes, independently of the witness size.

The Recursive Protocol

$\mathcal{P}(\mathbb{x}_0, \mathbb{w}_0)$:

1: **for** $i \in [\ell]$:
2: run one iteration on $(\mathbb{x}_i, \mathbb{w}_i)$;
3: $\mathbb{w}_{i+1} \leftarrow$ its output;
4: compute $\mathbb{x}_{i+1}$ as the verifier does;
5: *// last iteration*
send $\mathbb{w}_\ell$.

$\mathcal{V}(\mathbb{x}_0) \to \{0, 1\}$:

1: **for** $i \in [\ell]$:
2: run one iteration on $\mathbb{x}_i$;
3: $\mathbb{x}_{i+1} \leftarrow$ its output;
4: **receive** $\mathbb{w}_\ell$;
5: **return** $[\![(\mathbb{x}_\ell, \mathbb{w}_\ell) \in \mathcal{R}_{\text{dot}}]\!]$.

If any check inside an iteration fails, the verifier rejects. The proof consists of the short messages π of all iterations and the final witness $\mathbb{w}_\ell$.

Question #4

Suppose the witness has $N = 2^{21}$ ring elements. Show that $r \approx N^{1/3}$ minimizes $2N/r + r^2$, and compute the size of the new witness.

Parameters and Security

Norm Bounds

The honest prover must pass $\sum_i \|\mathbf{s}'_i\|_2^2 \leq \beta'^2$. LaBRADOR models all coefficients as independent Gaussians (matches experiments).

- Coefficients of $\mathbf{s}_i$: standard deviation $\sigma = \beta/\sqrt{rnd}$. Each coefficient of $\mathbf{z}$: $\approx \sigma\sqrt{r\tau}$, so

$$\|\mathbf{z}\|_2 \approx \sigma\sqrt{r\tau} \cdot \sqrt{nd} = \beta\sqrt{\tau} =: \gamma.$$

- $\mathbf{z}_{\text{low}}$ is essentially uniform modulo δ'' (std. dev. $\delta''/\sqrt{12}$), $\mathbf{z}_{\text{high}}$ has $\sigma\sqrt{r\tau}/\delta''$. Equal for

$$\delta'' \approx \sqrt{\sqrt{12r\tau} \cdot \sigma}.$$

- New bound: $\beta'^2 = \frac{2}{\delta''^2}\gamma^2 + \gamma'^2 + \gamma''^2$ (γ', γ'' bound the openings of $\mathbf{u}'$, $\mathbf{u}''$).

Longer-than-predicted honest vectors $\Rightarrow$ restart (running time, not soundness).

Concrete Parameters

- $\mathcal{R}_q = \mathbb{Z}_q[T]/(T^{64}+1)$ $(d = 64)$, $q \approx 2^{32}$ with $T^{64}+1$ splitting into two irreducible factors.
- Challenge set with $\tau = 71$ and $\theta = 15$.
- Commitment ranks κ, κ', κ'' chosen per iteration for 128 bits of Module-SIS security.
- Non-interactive via Fiat-Shamir; all challenges (including the projection matrices) are expanded from short seeds.

Reduction From R1CS

R1CS

Recall: $\mathbf{w}$ satisfies $(\mathbf{L}, \mathbf{R}, \mathbf{O})$ if

$$\mathbf{Lw} \odot \mathbf{Rw} = \mathbf{Ow}, \quad \mathbf{L}, \mathbf{R}, \mathbf{O} \in \mathbb{Z}_N^{k \times \ell} \quad (k \text{ constraints}, \ell \text{ variables}).$$

LaBRADOR supports $N = 2$ (*binary*: hash functions, bit manipulations) and $N = 2^d + 1$ (integer arithmetic). We discuss the binary case.

Binary R1CS. For $\mathbf{w} \in \{0, 1\}^\ell$, the prover computes

$$\mathbf{a} := \mathbf{Lw} \bmod 2, \quad \mathbf{b} := \mathbf{Rw} \bmod 2, \quad \mathbf{c} := \mathbf{Ow} \bmod 2,$$

packs all bits into polynomial coefficients and commits: $\mathbf{t} = \mathbf{A}_{\mathsf{R1CS}}(\mathbf{a} \,\|\, \mathbf{b} \,\|\, \mathbf{c} \,\|\, \mathbf{w})$.

It remains to show:

1: all committed coefficients are bits;
2: $\mathbf{a} \odot \mathbf{b} = \mathbf{c}$;
3: $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are computed correctly from $\mathbf{w}$ modulo 2.

Binarity and Products

Binarity. With $\mathbf{1}$ the all-ones vector of polynomials:

$$\mathbf{ct}(\langle \mathbf{a}^*, \mathbf{a} - \mathbf{1} \rangle) = \sum_i a_i(a_i - 1) \bmod q.$$

- $\mathbf{a}^*$ is not a dot product function of $\mathbf{a}$: add $\widetilde{\mathbf{a}} := \mathbf{a}^*$ to the witness; $\widetilde{\mathbf{a}} = \mathbf{a}^*$ is $\mathbb{Z}_q$-linear (permute, flip signs): constraints in $\mathcal{F}_{\mathsf{ct}}$.
- The constraint holds mod q, but we need it over $\mathbb{Z}$: every $a_i(a_i - 1) \geq 0$ and $\sum_i a_i(a_i - 1) \leq 2\|\mathbf{a}\|_2^2$. If the certified squared norm is below q, no wraparound: all a_i are bits.

Products. For bits, $ab = c$ iff $a + b - 2c \in \{0, 1\}$: show

$$\mathbf{ct}(\langle \widetilde{\mathbf{a}} + \widetilde{\mathbf{b}} - 2\widetilde{\mathbf{c}}, \mathbf{a} + \mathbf{b} - 2\mathbf{c} - \mathbf{1} \rangle) = 0.$$

Linear Relations Modulo 2

$3k$ relations modulo 2, but the main protocol works modulo q.
Compress them into λ random combinations:
$\boldsymbol{\alpha}_{i,L}, \boldsymbol{\alpha}_{i,R}, \boldsymbol{\alpha}_{i,O} \leftarrow\$ \{0,1\}^k$ for $i \in [\lambda]$, and

$$\boldsymbol{\delta}_i := \boldsymbol{\alpha}_{i,L}^\top \mathbf{L} + \boldsymbol{\alpha}_{i,R}^\top \mathbf{R} + \boldsymbol{\alpha}_{i,O}^\top \mathbf{O} \bmod 2 \in \{0,1\}^\ell.$$

The prover sends the integers (dot products over $\mathbb{Z}$ on the bits)

$$e_i := \langle \boldsymbol{\alpha}_{i,L}, \mathbf{a} \rangle + \langle \boldsymbol{\alpha}_{i,R}, \mathbf{b} \rangle + \langle \boldsymbol{\alpha}_{i,O}, \mathbf{c} \rangle - \langle \boldsymbol{\delta}_i, \mathbf{w} \rangle, \quad i \in [\lambda].$$

- All relations hold $\Rightarrow$ every e_i is even (the verifier checks).
- Some relation fails $\Rightarrow$ each e_i is odd w.p. $1/2$: soundness error $2^{-\lambda}$.
- Correctness of e_i: a constraint in $\mathcal{F}_{\mathsf{ct}}$, valid over $\mathbb{Z}$ as long as $\ell + 3k < q$.

The Binary R1CS Reduction

$\mathcal{P}(\mathbb{x} = (\mathbf{L}, \mathbf{R}, \mathbf{O}), \mathbb{w} = \mathbf{w})$:

1: $\mathbf{a}, \mathbf{b}, \mathbf{c} \leftarrow \mathbf{Lw}, \mathbf{Rw}, \mathbf{Ow} \bmod 2$;
2: **send** $\mathbf{t} \leftarrow \mathbf{A}_{\mathsf{R1CS}}(\mathbf{a} \,\|\, \mathbf{b} \,\|\, \mathbf{c} \,\|\, \mathbf{w})$;
3: **receive** $(\boldsymbol{\alpha}_{i,L}, \boldsymbol{\alpha}_{i,R}, \boldsymbol{\alpha}_{i,O})_{i \in [\lambda]}$;
4: compute $\boldsymbol{\delta}_i$ and e_i for $i \in [\lambda]$;
5: **send** $(e_i)_{i \in [\lambda]}$;
6: *// target witness*
return $(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{w}, \mathbf{a}^*, \mathbf{b}^*, \mathbf{c}^*, \mathbf{w}^*)$.

$\mathcal{V}(\mathbb{x} = (\mathbf{L}, \mathbf{R}, \mathbf{O}))$:

1: **receive** $\mathbf{t}$;
2: **send** $\boldsymbol{\alpha}_{i,L}, \boldsymbol{\alpha}_{i,R}, \boldsymbol{\alpha}_{i,O} \leftarrow\$ \{0,1\}^k$, $i \in [\lambda]$;
3: **receive** $(e_i)_{i \in [\lambda]}$;
4: **assert** e_i is even for all $i \in [\lambda]$;
5: *// target statement*
return $(\mathcal{F}, \mathcal{F}_{\mathsf{ct}}, \sqrt{2(3k + \ell)})$.

- $\mathcal{F}$: the opening of $\mathbf{t}$; $\mathcal{F}_{\mathsf{ct}}$: conjugation relations, binarity of $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{w}$, $\mathbf{a} + \mathbf{b} - 2\mathbf{c}$, and correctness of the e_i.
- Squared norm of the target witness $\leq 2(3k + \ell)$; $2(3k + \ell) \cdot \frac{128}{30} < q$ holds for all practical sizes with $q \approx 2^{32}$.

Summary

- LaBRADOR proves **dot product constraint systems** $\mathcal{R}_{\text{dot}}$: quadratic constraints over $\mathcal{R}_q$, constant-coefficient constraints, and a norm bound.
- Main protocol: **amortized opening** $\mathbf{z} = \sum_i c_i \mathbf{s}_i$, **garbage polynomials**, **aggregation** of all constraints into one, and a **JL projection** for the norm (slack ≈ 2.07).
- No zero-knowledge: no masks, no rejection sampling.
- The verifier's checks are again dot product constraints: **recursion** via outer commitments, decomposition and reshaping; the witness shrinks $N \to N^{2/3}$, so $O(\log \log N)$ iterations.
- Formally: a chain of **reductions of knowledge** that compose sequentially and in parallel.
- Binary R1CS reduces to $\mathcal{R}_{\text{dot}}$; 2^{20} constraints $\to \approx 58$ KB, from Module-SIS only.

Thank you for your attention



research.blockstream.com

github.com/BlockstreamResearch/zk-book



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